

Networking task 1

► What I've learnt!

► **1. IP Address**

It is an address of a device or a network.

2. MAC Address

It is a unique address ID for a local network.

3. Default Gateway

It is a device that our computers/phones send data to when the destination is outside our local network.

Usually, our Wi-Fi is the default gateway.

4. DNS (Domain Name System)

It converts website names into IP addresses.

Our computers communicate using IP addresses.

5. Difference between Public & Private IP

Private IP is used inside our local networks, e.g., home Wi-Fi.

► *** Public IP is the address our network uses on the internet.**

ipconfig

```
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Sanjana n>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::fd0d:dabd:beed:655b%4
    IPv4 Address. . . . . : 192.168.1.6
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\Sanjana n>|
```

The ping was successful

```
C:\Users\Sanjana n>ping google.com
```

```
Pinging google.com [142.251.222.206] with 32 bytes of data:
```

```
Reply from 142.251.222.206: bytes=32 time=14ms TTL=119
```

```
Reply from 142.251.222.206: bytes=32 time=15ms TTL=119
```

```
Reply from 142.251.222.206: bytes=32 time=15ms TTL=119
```

```
Reply from 142.251.222.206: bytes=32 time=14ms TTL=119
```

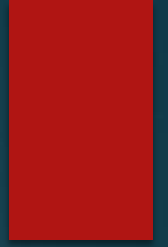
```
Ping statistics for 142.251.222.206:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 14ms, Maximum = 15ms, Average = 14ms
```

11 hops were shown, purpose is to understand the network path



```
C:\Users\Sanjana n>tracert google.com
```

```
Tracing route to google.com [2404:6800:4007:835::200e]  
over a maximum of 30 hops:
```

1	2 ms	5 ms	1 ms	2401:4900:1cbc:8ff6:324f:75ff:fede:758f
2	*	*	*	Request timed out.
3	*	6 ms	4 ms	2404:a800:3a00::4b9
4	16 ms	11 ms	12 ms	2404:a800::92
5	*	*	*	Request timed out.
6	537 ms	43 ms	50 ms	2404:6800:8201:80::1
7	15 ms	13 ms	12 ms	2001:4860:0:1::47de
8	46 ms	22 ms	14 ms	2001:4860:0:1::1840
9	12 ms	11 ms	11 ms	2001:4860:0:1::40e9
10	13 ms	11 ms	15 ms	2001:4860:0:1::565d
11	11 ms	11 ms	11 ms	pnmaaa-as-in-x0e.1e100.net [2404:6800:4007:835::200e]

```
Trace complete.
```

```
C:\Users\Sanjana n>
```

Basic network diagram

Device(client)
(Pvt IP or MAC)

Wifi router
(default gateway)

Modem
(from ISP)

ISP(internet service
provider)

The internet



Network information

SSID:	[REDACTED]
Protocol:	Wi-Fi 4 (802.11n)
Security type:	WPA2-Personal
Manufacturer:	Realtek Semiconductor Corp.
Description:	Realtek RTL8852BE-VS WiFi 6 802.11ax PCIe Adapter
Driver version:	6001.15.163.0
Network band (channel):	2.4 GHz (8)
Aggregated link speed (Receive/ Transmit):	150/150 (Mbps)
Link-local IPv6 address:	fe80::fd0d:dabd:beed:655b%4
IPv4 address:	[REDACTED]
IPv4 default gateway:	192.168.1.1
IPv4 DNS servers:	192.168.1.1 (Unencrypted)
Physical address (MAC):	[REDACTED]